Doubt Session

30 March 2026 20:12

Q: Convert a decimal number to binary number.

Eg:- num = 48

48 → binary

2	48	0
2	24	0
2	12	0
2	6	0
2	3	1
	1	1

2) 48 (24)
48
0

2) 24 (12
24
0

2) 6 (3
6
0

2) 3 (1
2
1

48 Binary → 110000
54 32 1

$2^5 + 2^4$
$= 32 + 16$
$= 48$ Ans.

Q: 16 to Binary Number:

2	16	0
2	8	0
2	4	0
2	2	0
	1	

10000

Q. 97 → to binary.

2	97	1
2	48	0
2	24	0
2	12	0
2	6	0
2	3	1
	1	

2) 97 (48
8
17
16
1

1 → 1100001

Code!

```
num = ip();
int ans=0;
while (num != 0){
→ int rem = num%2;
→ ans = ans*10 + rem;
→ num = num/2;
}
//rev the digit
```

→ Code

→ Quotient

ans = 1110

1 | 1 | 1 | 0

num = 8

0111

→ 0 111

rev the digit.

n = 8 , ans = 0 ; ans = ans × 10 + rem;

1) rem = 0
ans = 8×10 + 0
→ = 0

8 → 10000

2) n = 4
rem = 0
ans = 0×10 + 0
= 0

00001
→ 10000

1
0 1 ×

3) n = 2
rem = 0
ans = 0

ans = 1
→ rev 1

4) n = 1
rem = 1
ans = 0×10 + 1
= 1

n = 0

10000 00

1

8 = 10000

1×0 + 10×0 + 100×0 + 1000×0
+ 10000×1
= 10000

→ 4 → Thousands

8 = 10000

$1\times0 + 10\times0 + 100\times0 + 1000\times0 + 10000\times1 = 10000$

ones → 1 ×10
tens → 10 ×10
Hundreds → 100 ×10
Thousands → 1000 ×10
Ten'ss → 10000 ×10

4 → Thousands
1 → ones
4001

$4\times1000 = 4000 + 1 = 4001$

2	8	0
2	4	0
2	2	0
	1	

n=8

1 ×10 → 10 ×10 → 100 ×10 → 1000

1000 → 0001

$0\times1 + 0\times10 + 0\times100 + 1\times1000 =$

Decimal to Binary

```
int place_value = 1;  ans = 0;
while (num != 0){
    int digit = num % 2;
    ans = ans + digit × place_value;
    place_value = place_value ×10;
}
Sout(ans);
```

Q. Add two binary numbers → give the result in binary number.

Int range

$-2147\ldots$ to $2147\ldots$ (+1)

* Max_int 2,147,483,647

741 389 741 2

int rev = 0;

long rev = 0;

2L 3L 2L

10L → 10L

20L → >10L

if (rev > MAX_INT || rev < MIN_INT) return 0;

x^n double Power = 1;

$x \times x \times x \ldots \times x$

double Power = 1; n times

if (n<0) {
    n = -n; //Positive n

$n<0$

$x = \frac{1}{x}$

n = -n

$2^2 = 4$

$2^{-2} = \frac{1}{2^2} = \frac{1}{4}$

```
if (n < 0)
    n = -n;   //Positive n
    x = 1/x;
}
while(n != 0){
    Power = Power × x;
    n--;
}
return power;
```

$\boxed{n = -n}$

$n = 3$

$x = 2$

Power = 1

$\hookrightarrow 1 \times 2 = 2$

$n = 2$

$\hookrightarrow 2 \times 2 = 4$

$n = 1$

$\hookrightarrow 2 \times 2 \times 2 = 8$

$\hookrightarrow \boxed{n = 0}$

$7^4 - 4^7$

$$\boxed{x^{-n} = \frac{1}{x^n}}$$

task $x = 2.000$, $n = 16$

$\boxed{(2)^{16}}$

$\boxed{x = 2, n = 16}$ $\boxed{(x)^n = x^{n/2} \times x^{n/2}}$

$\hookrightarrow (x^2)^{n/2}$

$\hookrightarrow x^n$

$\rightarrow (2^2)^8$

$\rightarrow \boxed{(4)^8}$ $\boxed{x = 4 \quad n = 8}$

$\rightarrow (4^2)^4 = (16)^4$ $x = 16$, $n = 4$

$$\begin{array}{r} 32 \\ 32 \\ \hline 64 \\ 96 \\ \hline 4 \end{array}$$

$(32)^2$ $\boxed{x = 32}$, $n = 2$ $\boxed{n = 1}$

$(32^2)^1 \rightarrow (1024)^1$

$\boxed{32 \times 32}$ $\boxed{\text{Power} = 1}$ $\boxed{x = 1024}$

Power = Power × x

Power = 1024

$\boxed{2^5}$ $\rightarrow 2 \quad \boxed{2^4}$ $\boxed{x = 2, n = 4}$

$\boxed{x = x \times x, \; n = n/2;}$

Power = 1024

Power = 1×2

x = x×x
n = n/2;

$(4)^2$

x=4 n=2

odd

x=

x